DO NOW — RECALL

Lesson 063 · Thinking Logically: Conditions and Decisior

Name: _____ Class: _____ Date: _____

1. What does a Boolean expression evaluate to?

2. What is a decision boundary?

3. Why is condition order important in else-if?

4. Simplify (A AND B) OR (A AND NOT B), showing the Boolean laws used.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

True or False.

2. Model answer

A value at which condition outcome changes.

3. Model answer

Earlier branch can prevent later branch ever being checked.

4. Model answer

A AND (B OR NOT B) = A AND 1 = A. Use distribution, complement and identity.

By the end, you can…

• Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

• Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

• Use truth tables or decision tables to test normal, erroneous and boundary input values.

Vocabulary — know these terms

predicate — Boolean condition.

selection — Branch based on a condition.

boundary test — Input near comparison threshold.

short-circuit — Skipping redundant Boolean evaluation.

NEW LEARNING

Read, annotate and explain

Identify

Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

Model

Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

Evaluate

Use truth tables or decision tables to test normal, erroneous and boundary input values.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace Grade(40) using the conditions shown.

Explain the decision

Check comparisons in order; stop after first matching return.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace Grade for marks 39,40,69,70 and give output at each. [4]

Question 2

Write a Boolean rule for a valid age 11–18 inclusive, and give two boundary tests. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A program checks if(mark>=40) before else-if(mark>=70), then outputs Pass for 85. Explain bug and repair. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

With else-if, a later condition always runs even if an earlier branch matches.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.